

Executive Overview

ENERGY FORECAST & GRID INTELLIGENCE

Total Energy Consumption

1.25M kWh

Average Energy Consumption

25.03 kWh

Peak Demand

15 kW

Total Daily Consumption

2.08M kWh

Average Peak Demand

7.85 kW

Average Load Factor

58.92%

GRID PERFORMANCE

Average Grid Load

104.11 kW

Grid Capacity Utilization

34.70%

Average Line Loss

5.49%

Demand Forecast Error

94.36

Average Reserve Margin

17.52%

ENERGY & GRID TRENDS

Energy Consumption Trend

| Month    | Energy Consumption (kWh) |
|----------|--------------------------|
| Jan 2023 | 108,500                  |
| Feb 2023 | 93,000                   |
| Mar 2023 | 108,000                  |
| Apr 2023 | 107,500                  |
| May 2023 | 110,000                  |
| Jun 2023 | 96,500                   |
| Jul 2023 | 109,500                  |
| Aug 2023 | 109,500                  |
| Sep 2023 | 91,000                   |
| Oct 2023 | 102,500                  |
| Nov 2023 | 106,500                  |
| Dec 2023 | 107,500                  |

Actual vs Forecasted Grid Load by Region

| Grid Region | Actual Grid Load (kW) | Demand Forecast (kW) |
|-------------|-----------------------|----------------------|
| East        | 105                   | 175                  |
| North       | 105                   | 175                  |
| South       | 105                   | 175                  |
| Unknown     | 0                     | 175                  |
| West        | 105                   | 175                  |

Grid Load by Substation

| Substation | Average Grid Load (kW) |
|------------|------------------------|
| SS1        | 105                    |
| SS2        | 105                    |
| SS3        | 105                    |

Grid Load vs Capacity by Substation

| Substation | Average Grid Load (kW) | Average Grid Capacity (kW) |
|------------|------------------------|----------------------------|
| SS1        | 105                    | 300                        |
| SS2        | 105                    | 300                        |
| SS3        | 105                    | 300                        |

Energy Consumption by Device Category

| Device Category | Energy Consumption (kWh) |
|-----------------|--------------------------|
| Dryer           | 255,000                  |
| HVAC            | 255,000                  |
| Refrigerator    | 255,000                  |
| Washer          | 255,000                  |

Energy Consumption by Weather

| Weather Condition | Average Energy Consumption (kWh) |
|-------------------|----------------------------------|
| Cloudy            | 26,000                           |
| Rainy             | 26,000                           |
| Sunny             | 26,000                           |

Peak vs Off-Peak Rates by Tariff Plan

| Tariff Plan | Average Peak Rate | Average Off-Peak Rate |
|-------------|-------------------|-----------------------|
| Commercial  | 10.0              | 5.0                   |
| Industrial  | 10.0              | 5.0                   |
| Residential | 10.0              | 5.0                   |
| Unknown     | 8.5               | 4.2                   |

TARIFF & COST INTELLIGENCE

Average Unit Rate

6.50

Average Monthly Bill

1,102.47

Total Billing Amount

55.12M

Peak vs Off-Peak Rate Difference

4.98

Effective Cost per Unit

3.39